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Measurement of bone mineral density via light scattering

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Abstract

In this study we have investigated the potential of optical techniques to monitor changes in bone mineral density (BMD) via changes in scattering coefficient. For each of five bone samples, diffuse reflection and transmission coefficients were measured over the wavelength range 520–960 nm using an integrating sphere and CCD spectrometer. These were converted into optical absorption and scattering coefficients using a Monte Carlo inversion procedure. Measurements were made on samples immersed in formic acid solution for different lengths of time in order to investigate the effect of reduction in BMD on the optical properties. After full demineralization, the optical scattering coefficient fell by a factor 4. From the observed degree of fluctuation of the measurements, we estimate that BMD could be measured with an accuracy of 7% if optical scattering can be measured with an accuracy of 10%. We also report preliminary measurements of bone scattering using optical coherence tomography (OCT). An inter-side variability of 3% is obtained on dry samples with and without overlying periosteum. These results suggest that minimally invasive techniques for measuring optical scattering, such as OCT, may have a role in monitoring regional changes in BMD. This could be an important advance in our understanding of bone remodelling and its relationship to osteoarthritis. Both the integrating sphere and OCT measurements also suggest that light transport in bone is spatially anisotropic. OCT was used to assess probability of obtaining results *in vivo*.

1. Introduction

The incidence of osteoarthritis in the equine athlete, particularly in the joints of the fore and hind limbs, is relatively high at early stages of working life. The condition in the ridden horse is painful and associated with varying degrees of lameness. It is not uncommon for horses presented for clinical assessment to demonstrate signs of lameness in more than one limb.

It is generally accepted that osteoarthritic change in limb joints of ridden horses is the result of very vigorous exercise while carrying abnormal loads, particularly where the ground is 'hard' and the ground reaction force high. Osteoarthritis in the horse where degenerative changes in the joint cartilage are established is painful, irreversible, progressive and untreatable. The aetiology of osteoarthritis is not fully understood.

The equine limb joints of particular interest to us are the metacarpal–phalangeal of both fore and hind limbs. Current research at Exeter includes scintigraphic imaging and assessment of relative rate of sub-chondral and cancellous bone remodelling, digital imaging of section of the distal epiphysis of the III metacarpus taken post-mortem to demonstrate changes in trabecular patterns and localized differences in bone mineral density (BMD) and transport of nutrient to and within the joint cartilage. It has been hypothesized that changes in BMD of sub-chondral and cancellous bone of the epiphysis and cortical bone of the shaft of metacarpal III could initiate or be associated with osteoarthritic change as the magnitude of BMD can be related to the stiffness or energy absorbing property of the bone.

Scintigraphic imaging has been shown at Exeter to be a reliable indicator of the relative rate of sub-chondral and cancellous bone remodelling associated with osteoarthritic change and lameness (Attenburrow *et al* 1989). Images from 1000 horses showing signs of lameness have been obtained and of that number 247 were, subsequent to scintigraphic imaging, examined post-mortem. A close correlation between osteoarthritic change—degenerative changes in joint cartilage—and rate of bone remodelling is established.

Scintigraphic imaging used in the diagnosis and investigations in osteoarthritis although very useful can only, as applied to the horse, indicate relative differences in bone remodelling. However, such measurements together with absolute measurement of BMD could result in significant advances in the understanding of equine osteoarthritis.

A 'gold-standard' technique for measuring changes in BMD in humans exists, 'dual-energy x-ray absorption' (DEXA); however this technique is not suitable for application in the field (Speller *et al* 1989).

Cortical bone consists of an underlying matrix of collagen fibres, around which calcium-bearing hydroxyapatite crystals have been deposited. The principle of DEXA is that as BMD increases, the x-ray absorption by calcium increases due to the increased concentration of these crystals. However, these same crystals also cause bone to elastically scatter optical photons intensely. Initial work by Takeuchi *et al* (1997) indicates a correlation between optical attenuation and BMD as measured by DEXA, which could form the basis for a minimally invasive method of BMD determination, suitable for use *in vivo* and in the field. Specifically, these authors found that light penetration decreased with increasing BMD.

The aim of our study is to investigate the potential of minimally invasive optical techniques (e.g., optical coherence tomography (OCT)) (Swanson *et al* 1993) to monitor changes in BMD by detecting alterations in bone optical scattering. If sufficient accuracy can be demonstrated using such a technique, and a minimally invasive sensor constructed that is useable in the field, this could ultimately lead to a better understanding of the processes leading to osteoarthritis in the horse.

1.1. Definition of optical scattering coefficient

The scattering properties of a sample are described by a scattering coefficient μ_s . The physical meaning of the optical scattering coefficient can be understood by referring to the cuvette model for a solution containing a compound which causes multiple light scattering (with no absorption) as presented in figure 1. I_0 represents the incident light intensity and I is the intensity of unscattered light. The scattering coefficient μ_s is then defined via the relationship

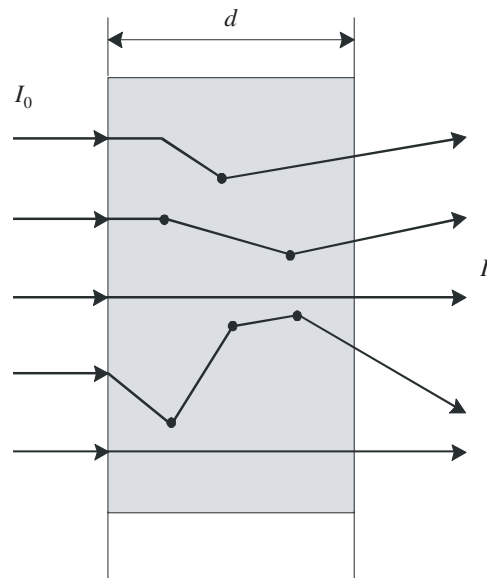


Figure 1. The cuvette model for a solution containing a compound that causes multiple light scattering (with no absorption). The scattering coefficient is defined in terms of the incident flux I_0 and unscattered transmitted flux I .

$I = I_0 \exp(-\mu_s d)$. A similar definition exists for the absorption coefficient μ_a , if I_0 and I refer to the incident and transmitted light for a cuvette containing a non-scattering, absorbing solution. The scattering coefficient for a medium μ_s containing a number density ρ of scatterers, each with total scattering cross section σ_s , is given as: $\mu_s = \rho \sigma_s$. Hence μ_s should be related to the density of hydroxyapatite crystals and thus to the BMD.

The scattering coefficient, and the absorption coefficient, of a tissue sample can also be determined from a measurement of the diffuse reflectance R and diffuse transmittance T of light by that sample. Such measurements can readily be made using integrating spheres. The measurement of the optical properties of different biological tissues in this way has been established since the beginning of the twentieth century (Taylor 1920, Jacquez and Kuppenheim 1955). We can use for our purposes two integrating spheres simultaneously (Pickering *et al* 1992) or make R and T measurements consecutively with only one integrating sphere (Yaroslavsky *et al* 1996, Simpson *et al* 1998). Both methods have advantages and disadvantages. For instance, if we use only one sphere with an inhomogeneous sample, we might obtain a false result if we measure transmittance and reflectance for different parts of the sample. If we use two spheres we have to pay attention to the fact that part of the incident light that went through the sample to the second sphere, can return to the first sphere, i.e., we ought to consider the interaction of two spheres. If we need to define optical properties of blood or living cells, the time of measurements is very important and we prefer to use two integrating spheres. However if our sample has permanent properties, one sphere is preferable, to avoid the influence of reflected light from the inside wall of the second sphere.

In order to convert measurements of diffuse reflectance R and transmittance T into estimates of μ_a and μ_s , it is necessary to have some model of light transport in tissue which relates these quantities. For our study a Monte Carlo model was used. Also, the Kubelka–Munk theory (Kubelka 1948, Wan *et al* 1981, Gemert and Star 1987) can be used to determine the optical properties of tissues, however it fails in media exhibiting a high scattering

anisotropy (Tuchin *et al* 1994), a distinguishing feature of biological tissues in the optical and near-infrared (van der Zee 1992).

2. Materials and method

2.1. Experiment 1: measurement by integration spheres technique

These measurements will follow the method reported by Firbank *et al* (1993) for the measurement of porcine skull bone.

To determine scattering and absorption coefficients as a function of wavelength we invert measurements of diffuse reflection and transmission. The steps are

- (1) We measure the diffuse reflection and transmission for a sample using an integrating sphere.
- (2) We use a Monte Carlo algorithm to calculate the expected diffuse reflectance and transmittance values for a particular set of scattering and absorption coefficients. The process is repeated for many values of absorption and scattering, in order to produce a look-up table.
- (3) For each pair of data values R and T determined experimentally from a tissue sample measurement and a given wavelength, we search the look-up table for the closest match. This yields the required scattering and absorption coefficients as a function of wavelength.

Step 3 requires an assumption to be made about the form of the single-scattering phase function for the tissue (Nickell *et al* 2000). In this study we have used a Henyey–Greenstein scattering phase function with a scattering anisotropy factor g of 0.93. This is within the range of reported values for porcine skull (Firbank *et al* 1993), which suggest g varies by only a few per cent around this mean value over the wavelength range 650–950 nm.

Using this technique, we have investigated, for the first time to our knowledge, the optical properties of bone samples taken from the leg of the horse. Since our samples (which include only the cortical part of the bone) have stable characteristics over time, we are able to use a single integrating-sphere technique. We have performed measurements on samples of equine bone, taken from the shaft of the third metacarpal. Samples were cut and ground into discs of thickness approximately 2 mm and diameter approximately 15 mm such that the parallel faces of the disc were orientated parallel to the surface of the shaft. In this way the samples contained the largest possible fraction of cortical bone. The bone sample was then mounted in a sample holder, fitted with glass windows and filled with saline. The bone samples were thus measured whilst immersed in saline, which produces an optimal match of refractive indices and ensures that the dominant Fresnel reflection loss occurs at the air/glass interface. This procedure simplifies the task of producing an acceptable transmission and reflectance standard sample. The transmission standard is simply a saline-filled sample holder without bone, whilst the reflection standard is a 99% reflective Spectralon[®] reflectance standard, with a glass coverslip attached to its front surface using a slick of water (Simpson *et al* 1998). These standards thus introduce essentially identical Fresnel reflection losses as the bone samples themselves.

The sphere design allowed use of the ‘comparison’ method for determining the reflectance or transmission of a sample. The construction of the integrating sphere provides four open ports arranged around the ‘equator’ and one at the ‘north pole’ i.e., at the apex of the sphere. Figure 2 shows a plan view of the sphere, looking down from above the ‘north pole’. A collimated illumination beam enters through an open port and strikes either the sample or reference standard placed in another port on the opposite side of the sphere. The Fresnel reflection from the front surface of the sample is caught by a light trap appropriately placed in

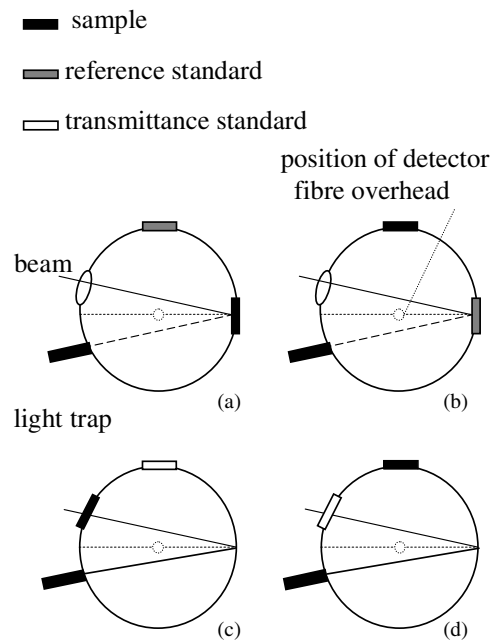


Figure 2. Integrating sphere measurement configuration used in these studies (view from above). Four measurements, reflectance and transmittance for the sample (a, c) and *ditto* for the standard (b, d) are required to characterize the absorption and scattering coefficients independently.

a third port as shown. The fourth equatorial port is occupied by the reference standard/sample. The port at the 'north pole' accommodates the detector fibre.

In the 'comparison' method, sample reflectance is measured by placing the sample in its holder, at an open port of a single integrating sphere and illuminating using an optical fibre-coupled white light source. The beam is collimated typically to a diameter of 5 ± 0.5 mm. A 99% reflective reference is placed in the second port, as shown in figure 2(a). The diffuse-reflected light is collected via a detector placed at the 'north pole' of the sphere and measured using a CCD spectrometer. To convert this detected intensity spectrum into reflectance, it is divided by the spectrum obtained by interchanging the sample and reference standard, as shown in figure 2(b). This procedure yields a 99% reflectivity spectrum obtained with the same average sphere reflecting as for the sample. A similar procedure is used to obtain the same transmittance spectrum. Measurements over all wavelengths from 520 to 960 nm could therefore be acquired simultaneously.

To estimate the repeatability of the integrating sphere system, we repeated measurements of the transmitted and reflected light by the same sample three times, each time fully replacing the sample holder and the light beam source. Variations of about 5% are observed in the measured diffuse reflectance and transmittance. This did not lead to significant differences in the corresponding derived values of the scattering and absorption coefficients. Likewise, freezing and then thawing the bone samples did not produce significant changes in the measured optical properties.

A gradual decrease in mineral content was produced by immersing the samples in formic acid solution for different lengths of time; this was to investigate the influence of a reduction in BMD on the optical properties. The degree of demineralization was quantified simply by

measuring the wet weight of the samples. The time for full demineralization was approximately 42 h for the sample with thickness 1.65 mm and much more for the other four samples whose thickness was 2 mm.

2.2. Experiment 2: measurements by OCT technique

Since our ultimate aim is to measure optical scattering *in vivo*, we have begun to study bone optical properties by OCT.

Our OCT measurements can be used to determine the single-scattering coefficient of an optically dense sample, given some knowledge about internal structure.

We used a bulk optics OCT system operating at wavelength of 1.3 μm to image various bone samples and to make images in different directions. A wavelength of 1.3 μm was chosen in preference to 1.5 μm so as to minimize the contribution of water absorption to the measurements. The power of the light source is 1 mW and the axial resolution is 18 μm . The lateral resolution is about 20 μm . To get knowledge about the influence of surrounding tissues we made measurements with and without the presence of the closest overlying layer of connective tissue—periosteum.

3. Results

3.1. Integrating sphere measurements

Plots of measured values for R and T versus wavelength are presented in figure 3. The black lines present the original sample whilst the grey lines show results after the sample had been immersed in formic acid solution, producing a maximal reduction in BMD. As expected R falls and T increases consistent with a reduction in elastic light scattering within the sample. The experimental results and a Monte Carlo method were used to obtain quantitative values for the unmodified scattering coefficient and absorption coefficient of the samples. Black and grey colours in figure 4 present corresponding values for μ_s and μ_a before and after full demineralization. Figure 4(b) shows that after full demineralization the recovered absorption coefficient appears to fall to zero beyond 760 nm. This result is spurious and probably results from a failure of the forward model to account fully for boundary effects at the sample and sample holder. These become important when long path photons are not excluded by absorption. However, our main interest is in the scattering coefficient and this appears to decrease linearly with bone mineral density, hence we believe our protocol is suitable for determining the degree of correlation between optical scattering and bone mineral density.

We have used the integrating sphere technique to determine μ_s as a function of bone mineral density for five samples taken from the legs of different horses. Our demineralization procedure using formic acid immersion caused the sample density to fall from 1.87 to 1.06 g ml⁻¹ (figure 5). The results of measurements showed the same influence of demineralization time on the reduction in BMD for all samples. Figure 6 shows that a full removal of calcium caused μ_s to fall at least by a factor 4 for all samples.

This shows that the calcium content in bone influences its optical properties considerably. However, it is also apparent that there is some variability of scattering between subjects, at the same nominal bone density (grey lines in figure 6). Clearly there is some other factor affecting the degree of scattering besides the density (as measured using weighing scales), the existence of which will affect the accuracy with which the optical scattering coefficient can predict the BMD.

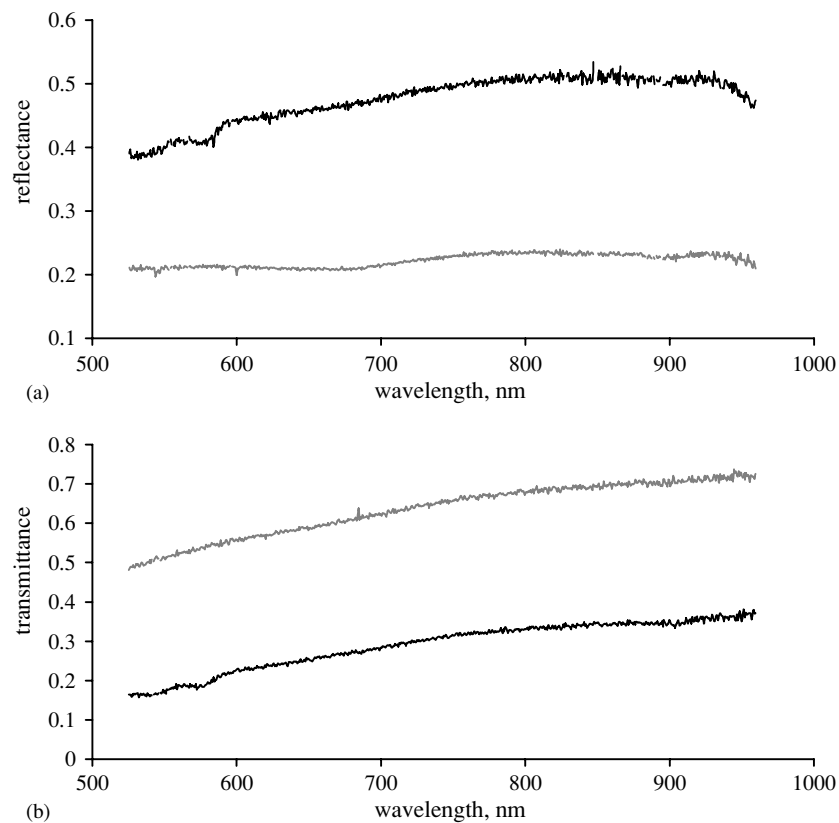


Figure 3. Plots of measured values of (a) diffuse reflectance R and (b) diffuse transmittance T of a bone sample before and after full demineralization.

The cause of the variation in the slope of the linear dependence of scattering coefficient versus BMD has not yet been fully understood. One hypothesis is that the orientation of the ground surfaces of our samples relative to the long axis of the bone shaft was not controlled accurately. This could lead to a different interaction between the light and the bone medium. For instance, it has been reported that the optical scattering coefficients of skin and muscle display pronounced differences depending on the direction in which the probing light travels, relative to the tissue fibre alignment direction (Nickell *et al* 2000). Since the mineralized collagen fibres in the bone shaft are predominantly aligned along the load-bearing direction, it is possible that our bone samples also have some directional anisotropy, related to the long-axis direction of the shaft. Since the direction-of-cut of our samples is not precisely defined, it may be that some of the variability seen in the last figure relates to this.

To test this hypothesis we have performed measurements in which the samples have been deliberately cut parallel, and at an angle (approximately 30°), to the surface of the shaft. As figure 7 shows, these two orientations produce different values for diffuse reflectance and transmittance, which in turn suggests a difference in optical scattering coefficient. Black and grey colours correspond to samples ground parallel and at an angle to the shaft, respectively.

The black lines (labelled 1 to 3) in figure 8 show the measured μ_s for the sample ground parallel to the shaft during three successive stages of demineralization. Likewise the grey lines (labelled 4 to 6) show the corresponding values for the samples ground at an angle. It is

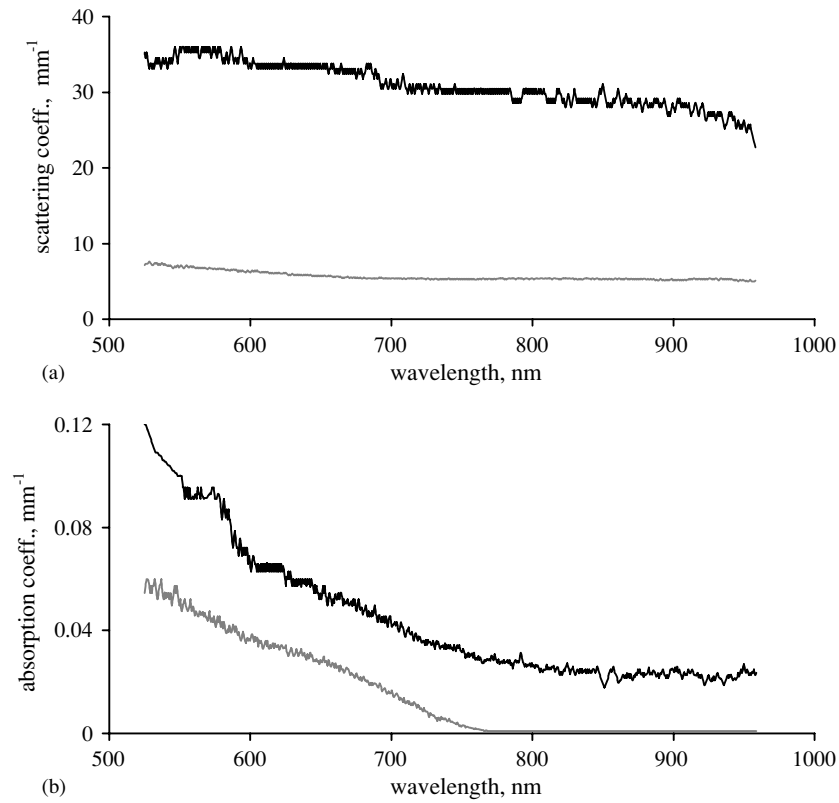


Figure 4. Plots for scattering coefficient (a) and absorption coefficient (b), for one-bone sample, before (black line) and after (grey line) full demineralization.

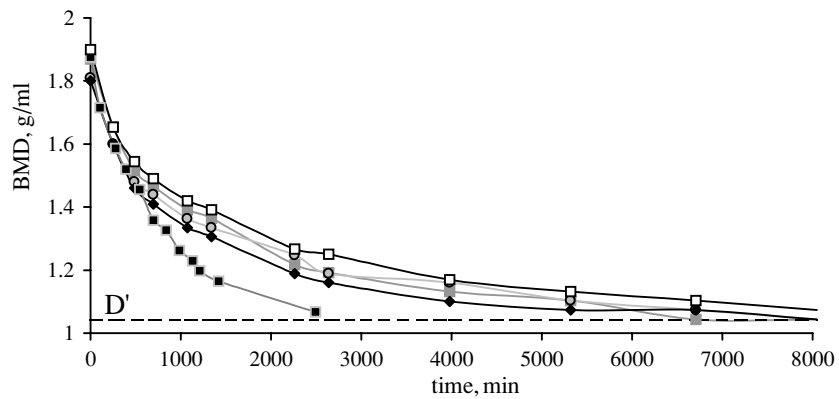


Figure 5. Reduction of BMD using formic acid for five different samples. BMD is inferred by measuring the wet weight (per ml) of the sample. D' —density of demineralized bone.

also apparent that, after full demineralization, the value of scattering coefficient depends much less strongly on the orientation of the grinding angle than the value prior to demineralization.

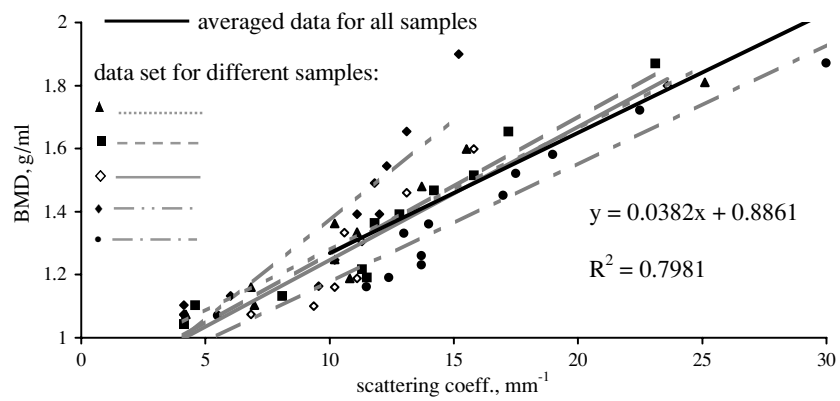


Figure 6. Correlation plots between BMD and optical scattering coefficient. Data pooled for all samples and for all demineralization times are represented by the black solid line. Grey lines and symbols show the correlation for each sample separately.

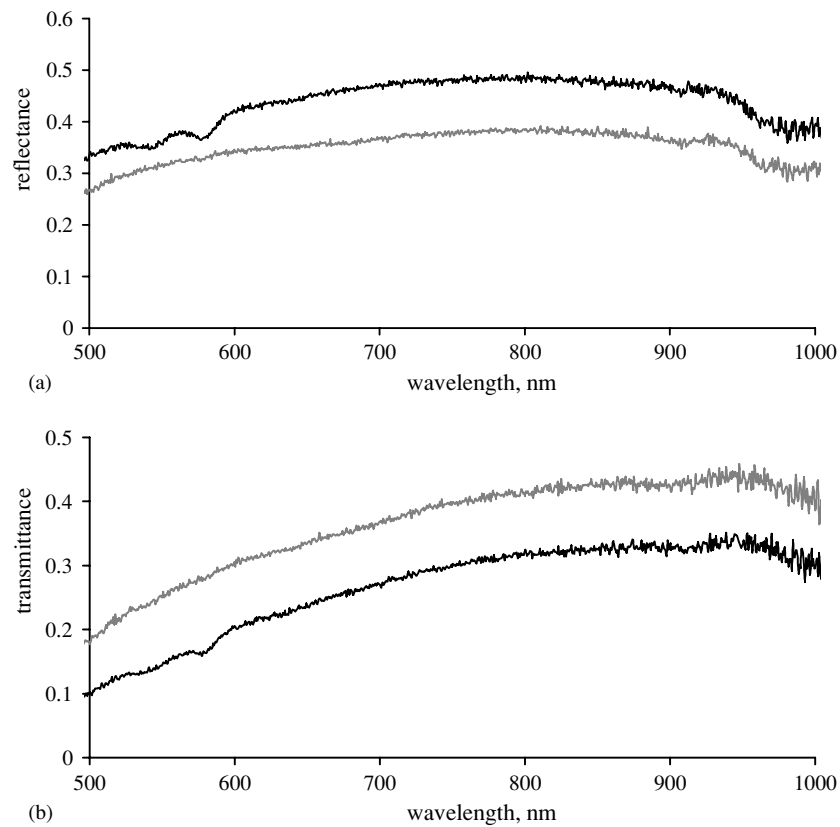


Figure 7. Plots of measured values of (a) R and (b) T for bone samples ground parallel (black line) and at 30° (grey line) to the long axis of the bone shaft.

This suggests that directional anisotropy of hydroxyapatite crystals is more important than directional anisotropy of collagen.

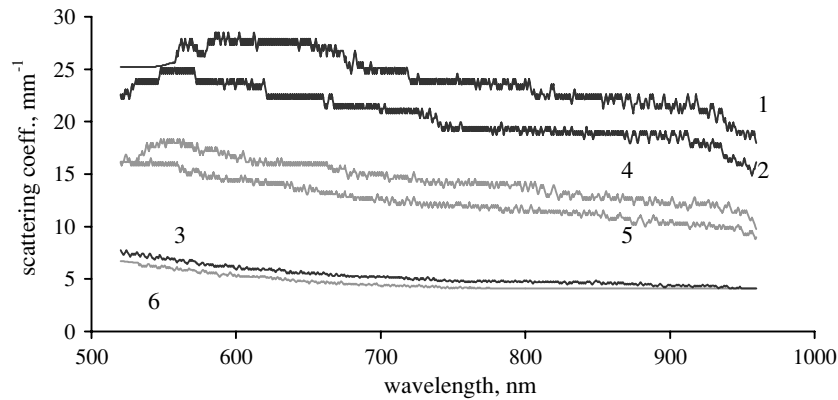


Figure 8. Variation in the scattering coefficient with demineralization time for samples cut parallel to the shaft long axis (lines).

To get further knowledge about cortical bone structure, we have carried out an experiment with two samples taken from the external and internal surfaces of the outer shell of cortical bone. The results are presented in figure 9. The R and T spectra of the external sample are presented by black lines. The grey lines show corresponding values of R and T from the internal layer. Whilst we have not explicitly inverted these data, it is clear that the scattering coefficient is higher (as R is higher and T lower) for the sample taken from the outer surface of the cortical shell.

3.2. OCT measurements

Figure 10 shows a typical OCT image of normal bone. The image format is 100×100 pixels. Note the grainy appearance, which is reminiscent of ultrasound images. This is ‘speckle’, a form of coherent noise that arises whenever partially coherent radiation is backscattered from a random medium such as tissue. By averaging over 100 neighbouring scans the speckle is averaged out (figure 11(a)).

The use of OCT to measure the single-scattering coefficient of biological samples has been described by Schmitt *et al* (1993). In its simplest form this method assumes that backscattered light from a sample decreases in intensity according to

$$I_{\text{backscattered}}(z) = I_0 \exp(-2(\mu_a + \mu_s)z)$$

where $\mu_a + \mu_s$ is the total attenuation coefficient and $2z$ is the round-trip distance of light backscattered at a depth z .

For most biological samples in the near-infrared μ_a (the absorption coefficient) is much less than μ_s (the unmodified scattering coefficient) hence μ_s can be estimated roughly as $\mu_s = 1/2z \ln(I_0/I(z))$ or as the gradient of a graph ($\ln I(z)$) versus z .

It is clear from a logarithmically scaled plot that a region exists over which the backscattered power decays exponentially (figure 11(b)). By performing a linear regression on this curve, the scattering coefficient can be determined (Schmitt *et al* 1993).

To investigate the accuracy of OCT measurements, we collected averaged images taken at different sites along the shaft of a bone sample. The measurement site was varied so as to keep the angle between the incident beam and the bone fibres constant. Figure 12 shows results of μ_s versus position for samples with (●) periosteum. An inter-site variability of 3% between the measured coefficients, determined by fitting the backscattered power curves, illustrates

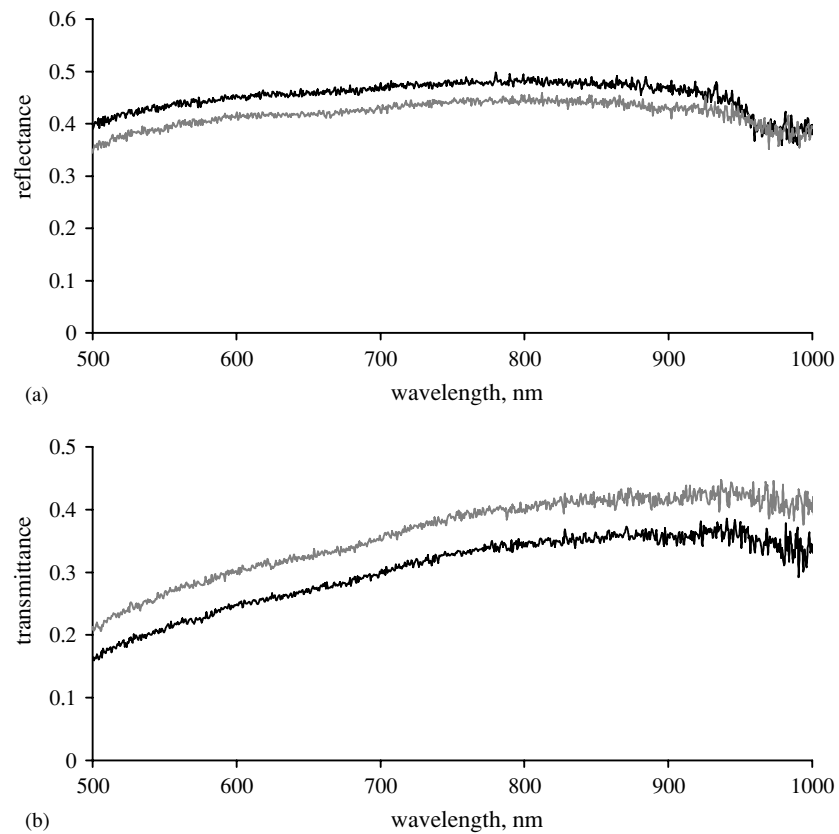


Figure 9. Plots of measured values of (a) R and (b) T for the samples from external and internal surfaces of the bone. Grey lines internal surface, black lines external surface of the cortical bone shell. It is evident that the optical scattering coefficient is higher when measured on the outer surface, suggesting a gradient of BMD within the shell with BMD increasing with radial distance from the centre of the shaft.

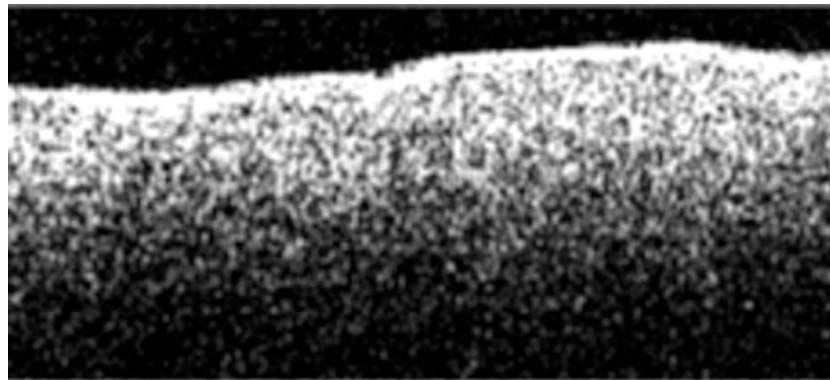


Figure 10. OCT image of normal bone with the incident beam perpendicular to the long axes of the shaft. The image format is 100×100 pixels.

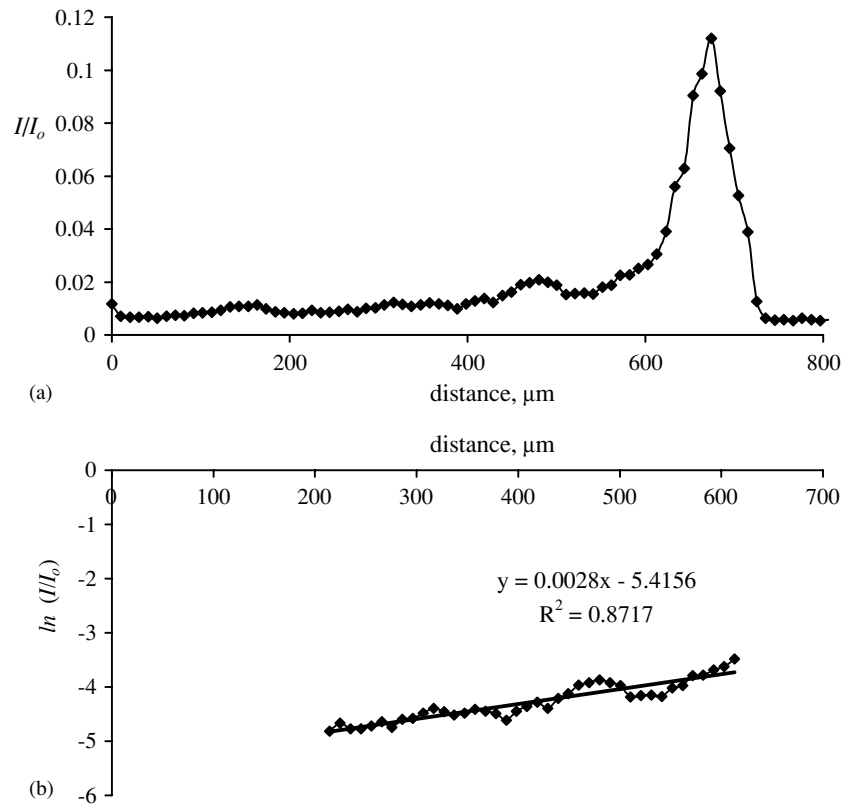


Figure 11. (a) The profile through this image by averaging over 100 neighbouring scans and (b) logarithmically scaled plot.

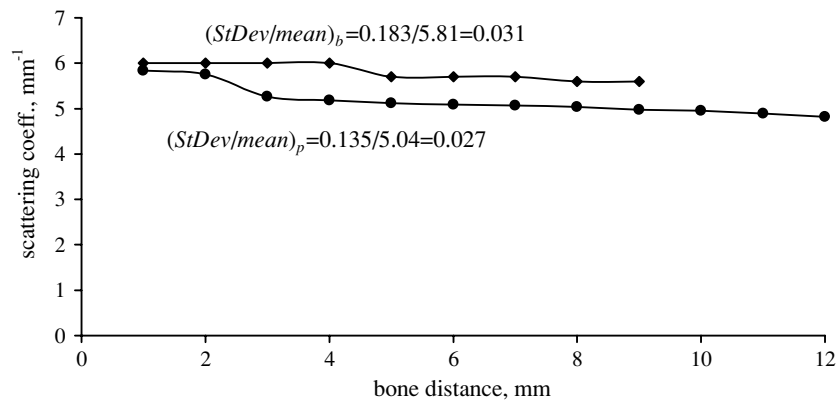


Figure 12. Plots of variation in the measured scattering coefficient along the long axis of a bone sample ($\blacklozenge$) in the absence and ($\bullet$) presence of overlying periosteum.

the likely accuracy of the OCT measurements when 100 lateral scans can be acquired and averaged, thereby reducing speckle noise. The small angle between sample surface and light beam might be responsible for the slight trend in the measured coefficients. Our original idea

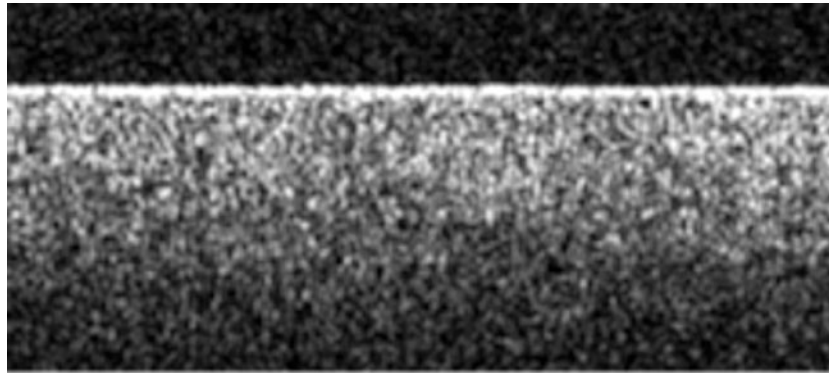


Figure 13. As for figure 10 but with the direction of the incident beam now parallel to the long axis of the shaft.

for *in vivo* measurements precludes such averaging however, as *in vivo* measurements must be made via a non-imaging ‘needle’ probe and so we are currently working both to assess the impact that un-averaged speckle noise will have on the measurement accuracy and to realize needle probes that will allow spatial averaging to be performed.

To assess the influence of the overlying layer of periosteum, measurements along a straight line parallel to the shaft long axis were performed for bone in which the surface periosteum had not been removed (figure 12, without (◆) periosteum). The similarity of the values suggests that the optical properties of dry periosteum are at most 20% different from those of the underlying bone. That is possible, due to presence of the same cells both in periosteum and in the calcified bone structure. Alternatively, it may indicate that the dry periosteum is so thin that it is not resolved by our OCT system. The periosteum signal would then be indistinguishable from the surface Fresnel reflection peak, hence excluded from our regression analysis.

To check our earlier observations about anisotropy of bone tissue, we measured by OCT, the scattering coefficient for bone by shedding the light beam perpendicular and parallel to the main direction of fibres. Corresponding images are presented in the figure 10 ($\mu_s = 2.8 \text{ mm}^{-1}$) and figure 13 ($\mu_s = 2 \text{ mm}^{-1}$). Changes in scattering coefficient about 40% confirm our previous suggestion concerning the directional anisotropy of bone, which is related to the long-axis direction of the shaft.

4. Discussion and conclusions

We have used a combined integrating sphere technique and Monte Carlo method to measure the optical coefficients (scattering and absorption) for samples of cortical bone take from the third metacarpal of the horse *ex vivo*. This study was motivated by our interest in determining BMD in competition racehorses *in vivo*, using a minimally invasive technique such as OCT. Our goal is to elucidate the relationship between cortical bone remodelling and the destruction of articular cartilage that characterizes osteoarthritis.

Our study extends earlier work by Takeuchi *et al* (1997) wherein it was demonstrated that light penetration increased with progressive demineralization. However, the authors only performed a preliminary analysis of the relative changes in absorption and scattering, using time-resolved spectroscopy at a single wavelength. In the present study, we used a technique that explicitly separates the scattering and absorption coefficient, and obtained quantitative measurements of these coefficients over the wavelength range 500–900 nm.

By demineralizing the bone samples in formic acid solution, we also demonstrated a significant linear correlation between the optical scattering coefficient and bone mineral density. Based on the measured inter-sample variability evident in figure 6, and the corresponding uncertainties in the regression parameters relating BMD to optical scattering, we have demonstrated by measurements *in vitro* that a potentially useful correlation exists between BMD and optical scattering coefficient. We can estimate that a measurement of optical scattering coefficient made with 10% accuracy should yield an estimate of BMD with an accuracy of better than 7%.

We have also made measurements using OCT, which can infer the total optical attenuation coefficient $\mu_t (= \mu_a + \mu_s)$ from measurements of the decay of coherently backscattered light flux as a function of depth. By making measurements at 1.3 μm , where $\mu_a \ll \mu_s$, this yields an estimate for μ_s . An inter-site variability of 3% suggests OCT may have a useful level of measurement accuracy, provided speckle averaging is performed. Since optical coherence tomography is compatible with fibre optics, this may pave the way for a minimally invasive assessment of localized changes in BMD in living horse.

The accuracy of the observed correlation is, in part, limited by the observed variability between bone samples of the regression line between μ_s and BMD. Initial observations suggest that some of this variability may arise because of variations in the angle at which the samples were ground (into disks) relative to the long axis of the bone shaft for the integrating sphere measurements and in the angle between the incident light and the main direction of the fibres for the OCT measurements. In the cortical bone the calcium hydroxyapatite crystals are believed to be a major source of light scattering. These crystals are deposited on an underlying network of collagen fibres, whose orientation closely follows the direction of the shaft long axis. The apparent optical properties of the bone are therefore likely to be spatially non-isotropic, as is observed for other filamentous tissues such as muscle and skin (Nickell *et al* 2000, Marquez *et al* 1998, Zijp Jaap and ten Bosch Jaap 1998). A maximal effect should be seen if samples cut at 0 and 90° are compared. In the latter sample, light 'guiding' along the fibre direction may occur, which intuitively may enhance transmission and suppress back scatter (i.e., reflectance). This would be interpreted by our inversion algorithm as a lowered scattering coefficient. It is interesting to note that the sample cut at an intermediate angle (30°, figure 7) does indeed show a lower apparent scattering coefficient than that cut at 0°. The same observation was indicated from OCT images. The results obtained from different measurement techniques strongly suggest that cortical bone tissue is an anisotropic tissue with a preferential direction related to the collagen fibre orientation. Optical anisotropy should also manifest itself as optical birefringence or dichroism. We are currently working to characterize the polarization properties of bone samples using polarization-sensitive OCT (Matcher *et al* 2003).

Acknowledgments

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